extensions

⚠️ **Security Warning**: Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

📋 Challenge Description

This is a really weird text file. Can you find the flag?

💡 Hints

- How do operating systems know what kind of file it is? (It's not just the ending!)
- Make sure to submit the flag as picoCTF{XXXXX}

🧰 Tools Used

- `file` – Identify file type
- `strings` – Extract printable character sequences
- `exiftool` – Read metadata
- `cp` – Copy / rename files

🔍 Complete Solution

Step 1: Download and Prepare

Download the file named `flag.txt` from the challenge page.

Step 2: Basic File Inspection

First, verify the file type:

```
file flag.txt
```

Expected output:

```
flag.txt: PNG image data, 1697 x 608, 8-bit/color RGB, non-interlaced
```

> **Key finding**: Even though the file is named `flag.txt`, the `file` command reveals it's actually a **PNG image**! Operating systems identify file types by **magic bytes** (file signatures), not just file extensions.

Step 3: Search for Visible Text

Check if any readable text reveals the flag:

```
strings flag.txt | grep -i "picoCTF"
```

Output: (no results)

No flag yet, but we know it's an image now.

Step 4: Examine Metadata

Use `exiftool` to inspect the file's internal metadata:

```
exiftool flag.txt
```

Output:

```
ExifTool Version Number         : 13.50
File Name                       : flag.txt
Directory                       : .
File Size                       : 10.0 kB
File Modification Date/Time     : 2026:05:20 12:12:50+02:00
...
File Type                       : PNG
File Type Extension             : png
```

```
MIME Type                       : image/png
Image Width                     : 1697
Image Height                    : 608
Bit Depth                       : 8
Color Type                      : RGB
Compression                     : Deflate/Inflate
Filter                          : Adaptive
Interlace                       : Noninterlaced
...
Image Size                      : 1697x608
Megapixels                      : 1.0
```

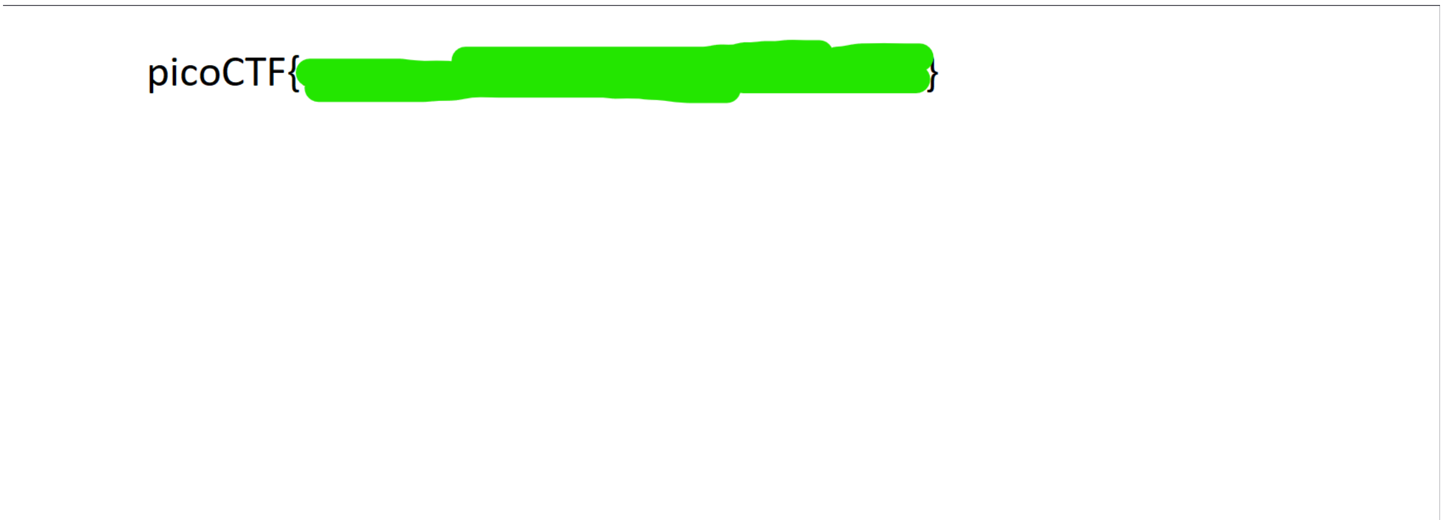
The metadata confirms this is a valid PNG image with dimensions 1697×608 pixels.

Step 5: Rename and Open the Image

Since the file is actually a PNG, rename it with the correct extension:

```
cp flag.txt flag.png
```

Now open the image:



The flag is visible directly in the image!

📌 Alternative Approach

Instead of renaming, you can also open `flag.txt` directly in an image viewer that reads file signatures rather than extensions (most modern viewers do this automatically).

✅ Final Result

```
picoCTF{<REDACTED>}
```

> **Note**: The actual flag is redacted here. In the real challenge, opening the renamed PNG image will reveal the complete flag.

🧠 Key Concepts

Concept	Description
Magic Bytes (File Signatures)	Unique byte sequences at the start of a file that identify its format, independent of file extension.
File Extension vs File Type	The `.txt` extension doesn't always mean a file is plain text. Always use `file` to verify.
PNG Signature	Valid PNG files start with the bytes `89 50 4E 47 0D 0A 1A 0A` (`‰PNG\r\n\x1a\n`).
exiftool	A tool that reads metadata and can identify file types even with incorrect extensions.

💡 Pro Tips

1. **Never trust file extensions** – Always use the `file` command to determine the true file type.
2. **The `file` command reads magic bytes** – It looks at the file's header, not its name, to identify what it really is.

3. **If `file` says it's an image, rename it** – Changing `.txt` to `.png`, `.jpg`, or `.gif` often reveals the flag.

4. **`exiftool` is great for metadata** – It can confirm the file type and show dimensions, color depth, and other details.

5. **Common misdirections** – CTF challenges often rename files (e.g., `flag.txt` is actually a PNG, `image.jpg` might be a ZIP). Always run `file` first!